

Robustness, Enhancement, and Fine-Tuning of Computer Vision Algorithms

Course: CS 3002 (Image Processing / Vision)

Datasets: COCO-2017 & ADE2K

Authors: Computer Vision Research Group

Evaluating algorithm robustness, restoration, and adaptation under environmental degradation.

Low-Level Vision

Feature Detection evaluated via ORB (Oriented FAST and Rotated BRIEF).

High-Level Recognition

Object Detection evaluated via YOLOv8n.

Dense Scene Understanding

Semantic Segmentation evaluated via DeepLabV3-ResNet50.

Quantifying performance degradation under Gaussian Noise, Salt & Pepper Noise, and Motion Blur, mapped against Signal-to-Noise Ratio (SNR).

The 6 Pillars of Experimental Architecture

Datasets COCO-2017 (80 object classes) & ADE2K (150 semantic categories).	Vision Tasks Keypoint Detection, Object Detection, Semantic Segmentation.	Metrics Keypoint Match Ratio, Per-Class Recall, mIoU, SNR (dB).
Models ORB, YOLOv8n, DeepLabV3-ResNet50.	Stressors Gaussian Noise ($\sigma \in [0, 100]$), Salt & Pepper ($p \in [0.05, 0.50]$), Motion Blur ($k \in [1, 29]$).	Mitigations Bilateral Filter, Median Filter, Restoration Filter / Deep Learning Fine-Tuning.

Establishing the Clean Baseline

Performance under ideal conditions (Control SNR: 50.00 dB).

ORB

1.000

Match Ratio | 785.2
Average Keypoints

YOLOv8n

1.000

Mean Recall | (Across
all evaluated bounding
box classes)

DeepLabV3

1.000

mIoU | (Relative to
clean baseline masks)

System Breakage Under Extreme Distortion

Distortion Type	Worst-Case SNR	ORB Match Drop	DeepLabV3 Drop
Gaussian Noise ($\sigma=100$)	SNR 4.59 dB	Drops to 0.252	Drops to 0.279
S&P Noise ($p=0.50$)	SNR 3.04 dB	Drops to 0.083	Drops to 0.111
Motion Blur ($k=29$)	SNR 15.50 dB	Drops to 0.054	Drops to 0.409

Class-Specific Sensitivity in Object Detection (YOLOv8n)

Tracking recall failure: Large structural targets vs. fine geometric objects.

Class: Person (Macro Structure)		Class: Cup (Fine Geometry)
0.941	SNR 22.50 dB	0.865
0.636	SNR 9.33 dB	0.514
0.305	SNR 4.59 dB	0.081
0.343	S&P Noise Terminal Point SNR 3.04 dB	0.000

Fine structures experience total catastrophic failure (0% recall), while macro structures retain partial resilience.

Mitigation Benchmark: Pre-Filtering vs. Fine-Tuning

YOLO S&P Noise ($p=0.35$)

Raw Input:
0.284

Pre-Filtered:
0.609

Fine-Tuned:
0.623

Winner:
Fine-Tuning
(+32.5% Gain)

DeepLabV3 Gaussian ($\sigma=70.0$)

Raw:
0.517

Filtered:
0.620

Fine-Tuned:
0.466

Winner: Bilateral Filter
(+10.3% Gain)

DeepLabV3 S&P Noise ($p=0.35$)

Raw:
0.127

Filtered:
0.644

Fine-Tuned:
0.519

Winner: Median Filter
(+51.7% Gain)

Strategic Verdict

1 Class-Specific Sensitivity

Fine geometric structures (cups, cars) fail drastically under noise, while large macro targets (people, chairs) retain significantly higher structural resilience.

2 Complementary Mitigations

Classical spatial filtering (Median/Bilateral) dominates noise suppression; deep neural network fine-tuning dominates motion blur recovery.

3 Absolute Reproducibility

All data points, measurements, and benchmark grids are fully deterministic and executable via `generate_actual_metrics_and_plots.py`